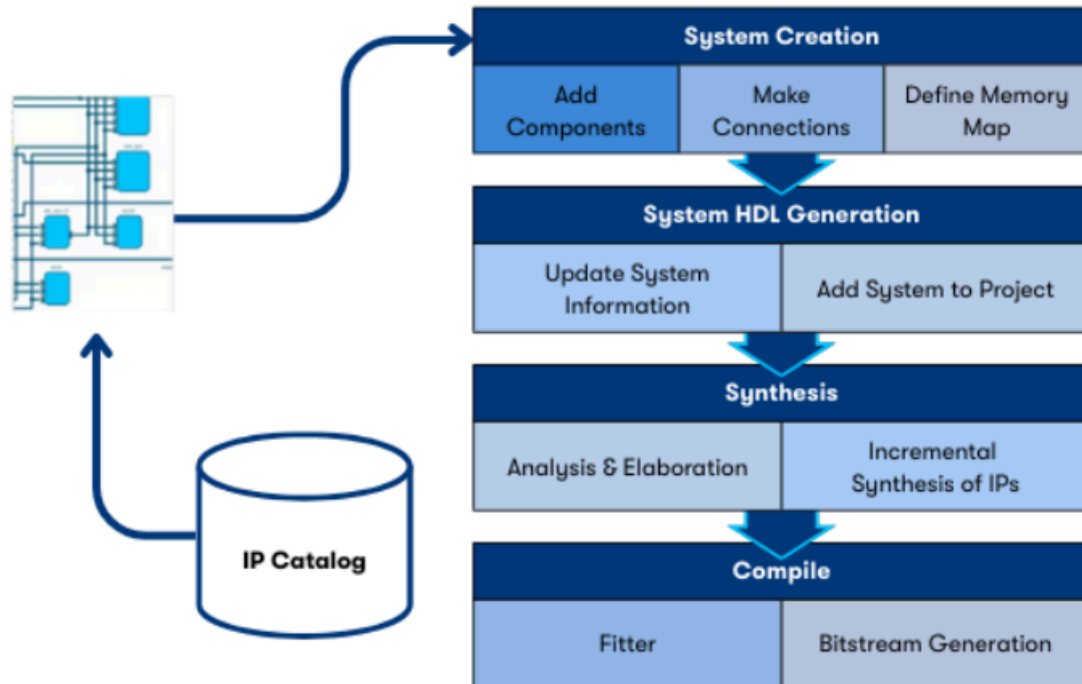


Visual Designer Studio



Supported interfaces

- Altera [Avalon](#) (AXI like)
 - Memory mapped
 - streaming
- ARM AMBA AXI
 - AXI3
 - AXI4
 - AXI4Lite
 - AXI4Stream
 - AXI4NoC
 - AXI5
- Conduit
 - directly exported, no interaction with connectivity
- Clock
 - Specify frequency
 - AXI and Avalon interfaces have associated clock (used to introduce clock crossing where required)
- Reset

- has associated clock
 - AXI and Avalon interfaces have associated reset (used to introduce reset synchronisation where required)
 - Interrupt
-

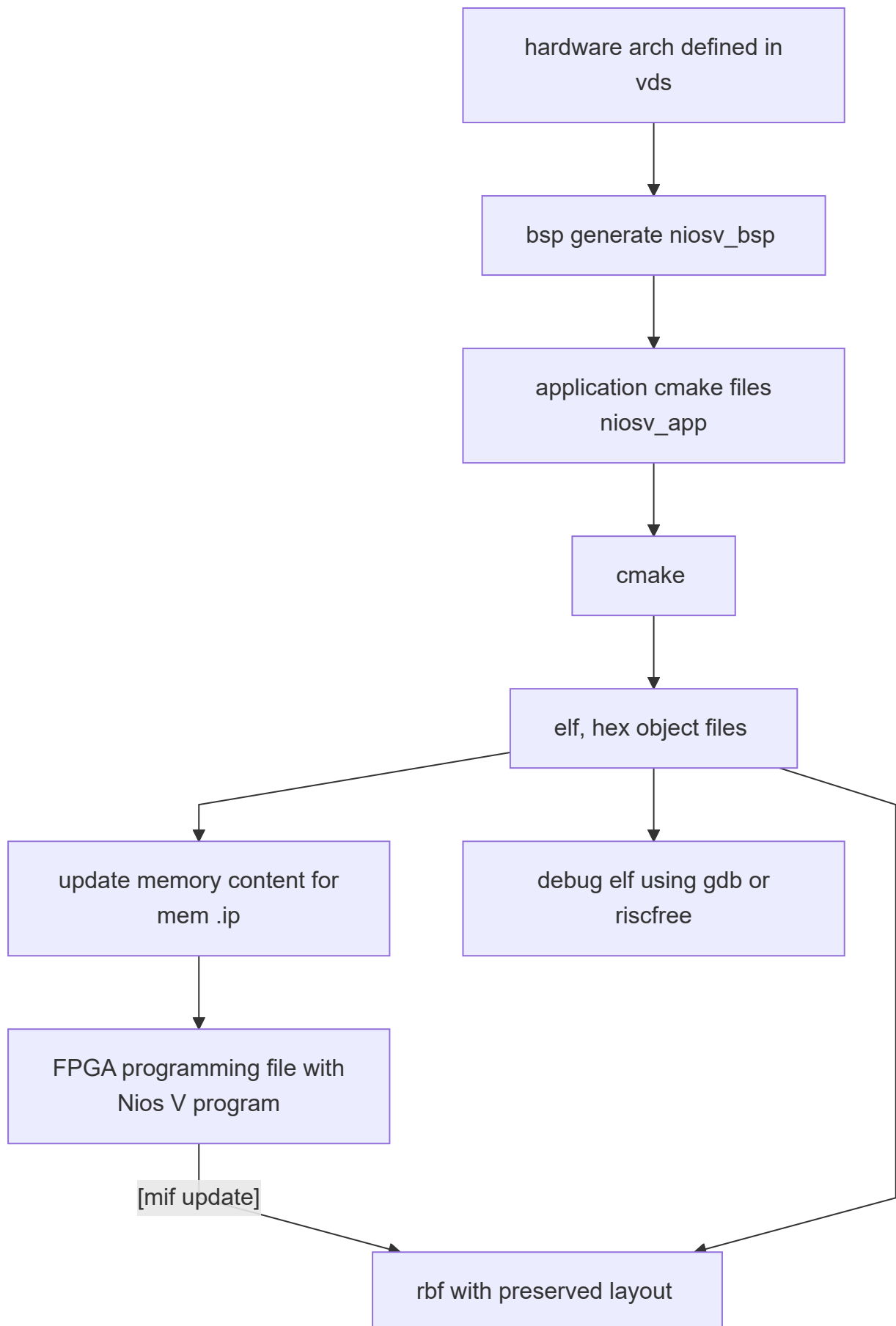
Interconnect generation

- Memory mapped interconnect
 - Any master to any slave
 - interface adaptation inferred
 - associated clock and reset
 - Option to explicitly insert components recommended
 - clock crossing bridge
 - width adaption
 - burst adaption
 - Variable pipelining at system level to trade latency vs frequency
 - Streaming
 - Any master to any slave
-

Software integration

::: notes

The columns renders left to right in the reveal.js generated by pandoc



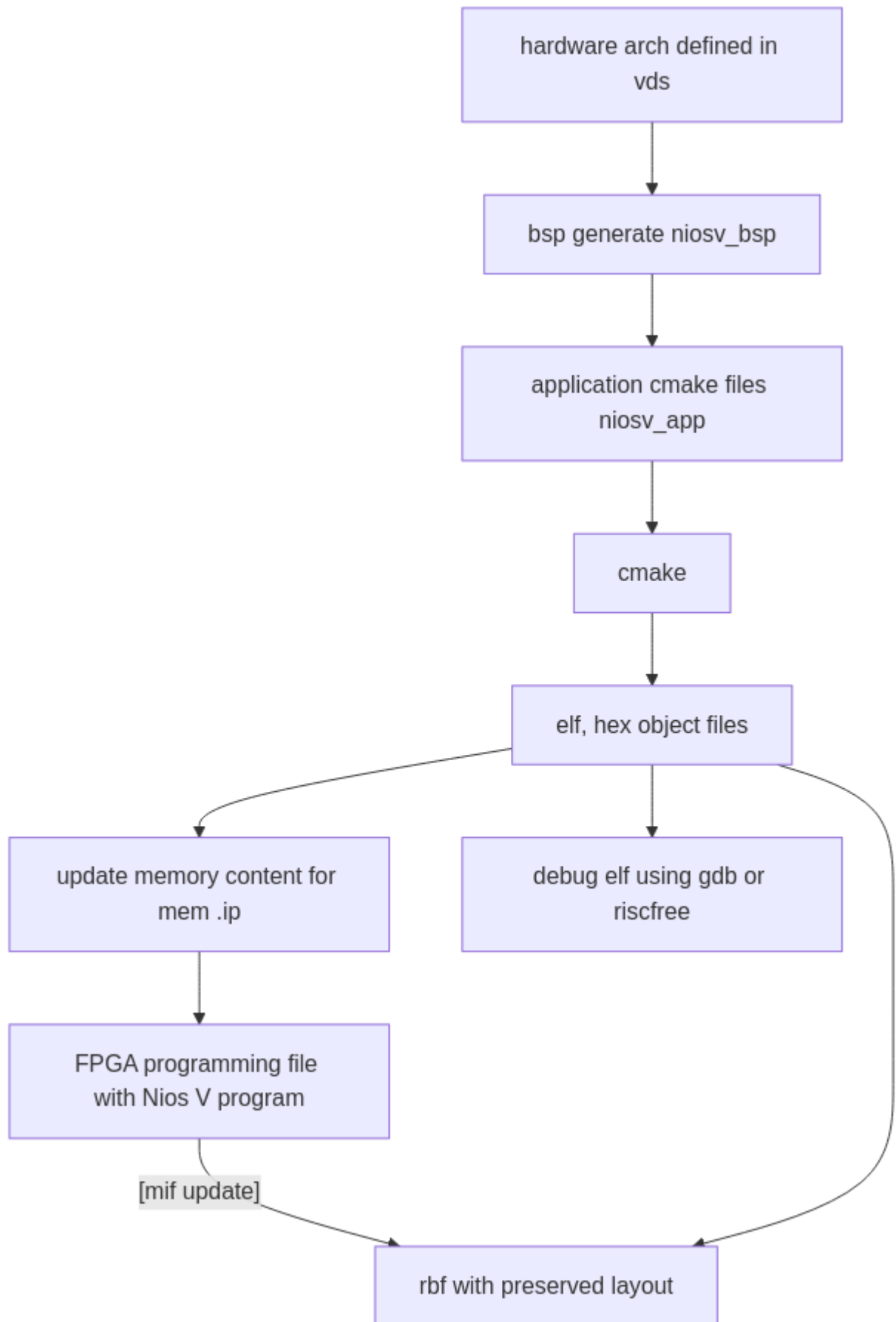
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- BSP editor `niosv-bsp` and application template editor `niosv-app` available as cmd line applications or GUI for BSP editor.
 - Generates cmake based build files
 - Optional RiscFree IDE available

:::

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...

...

Tour of the tool

⋮ notes

Better to do this part live, but cover

- IP catalog
 - Same view as you see in the Quartus Prime Pro window
- Canvas view - how to connect up
 - We cover more in the lab
- Connectivity Designer
 - Alternate to the canvas - looks like SOPC Builder and Platform Designer to people who have used those before.
- Address Map Viewer
- Interrupt Editor
- Subordinate Arbitration Editor
- Netlist Navigator / Connections window
- Property Window
- Domains window
 - Look at how interconnect constraints can be defined

⋮